

# Willian Vieira PhD

Geospatial machine learning engineer with a PhD in quantitative ecology. I build data systems that turn messy empirical data into reproducible simulations, transparent infrastructure, and decision-ready analytical products.

## DATA SCIENCE & ENGINEER EXPERIENCE

2025  
|  
2024  
(1 yr 9 m)

### • Data Analyst

**Habitat**, Montreal, Canada

Designed cloud data infrastructure on Azure for heterogeneous geospatial data, metadata-driven access, and feature retrieval for ML training and inference | Architected end-to-end R/Python ML pipelines to process and model large-scale, noisy real-world data for production analytics | Built metadata-driven ETL and provenance workflows to make datasets, models, assumptions, and analyses traceable, reproducible, and reliable | Led the transition to a Unix-based production environment using Docker, CI/CD, automated testing, and DevOps best practices | Pushed ML models into production, supporting client-facing data products with validation checks and monitoring

2024  
|  
2017

### • PhD Research

**Integrative Ecology Lab**, Sherbrooke, Canada

Built reproducible pipelines for continental-scale geospatial and environmental data, harmonizing climate, satellite, and field-inventory datasets across North America | Developed Bayesian hierarchical models and simulation workflows to represent heterogeneous forest populations, forecast dynamics, and propagate uncertainty from noisy, biased, incomplete real-world data | Ran thousands of simulations on HPC infrastructure () to test model behavior, compare assumptions, and scale analyses beyond local compute | Developed open-source software libraries (, ) for spatial demographic modelling with version control, automated testing, and reusable workflows | Published a **technical methods book** documenting the full computational pipeline, along with one peer-reviewed **publication** and two preprints (, )

2022  
|  
2020  
(part-time)

### • Biostatistician

**Environment and Climate Change Canada** - Quebec, Canada

Developed a cost-aware probability sampling protocol to improve the spatial representativeness of boreal bird surveys in Quebec | Led R&D on spatial bias-correction methods for large-scale ecological monitoring data, designing an approach later adopted by other provinces | Engineered a fully automated and reproducible **data pipeline** with version-controlled workflows, automated documentation, and stakeholder-ready analytical outputs

## EDUCATION

2024  
|  
2017

### • PhD, Ecology

**Université de Sherbrooke** - Sherbrooke, Canada

 *How climate, competition, and forest management shape the limits of tree species distributions: from individuals to metapopulations*

2016  
|  
2015

### • Masters 2, Agroecology and Resource Management

**Bordeaux Sciences Agro**, Bordeaux, France

 *Modelling the dispersion of weed species in agricultural landscapes*

2015  
|  
2010

### • BSc in Agronomy

**Universidade Federal de Santa Catarina**, Florianópolis, Brazil

## CONTACT

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 [LinkedIn/WillVieira](#)

 [GitHub/WillVieira](#)

 [Publications](#)

## SKILLS

### ML & Statistics:

Bayesian inference · Hierarchical models · Simulation modelling · Spatial modelling · Uncertainty quantification · Model evaluation

### Programming & Database:

Python · R · SQL · Bash · DuckDB · Stan · Julia

### Data Engineering & Cloud:

ETL/ELT · Geospatial data products · Cloud data infrastructure · APIs · Docker · Azure · AWS · GeoParquet · COGs · Apache Arrow · HPC

### DevOps & Reproducibility:

Git · CI/CD · Unix · Unit testing · Reproducible pipelines · Automated validation · Metadata · Provenance · Make

## OUTREACH

· Developed and maintained the **QCBS R Workshop Series**, reaching nearly one thousand graduate students · Taught programming, statistics, and ecology to 250+ students across biology, engineering, and graduate programs · Published **5 papers & preprints** with ~100 citations · Translated stakeholder needs into technical analyses, reproducible workflows, and clear analytical outputs

## LANGUAGES

Portuguese · Native  
French · Full Professional  
English · Full Professional

 [Source](#) ·  [HTML](#) ·  [PDF](#)

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